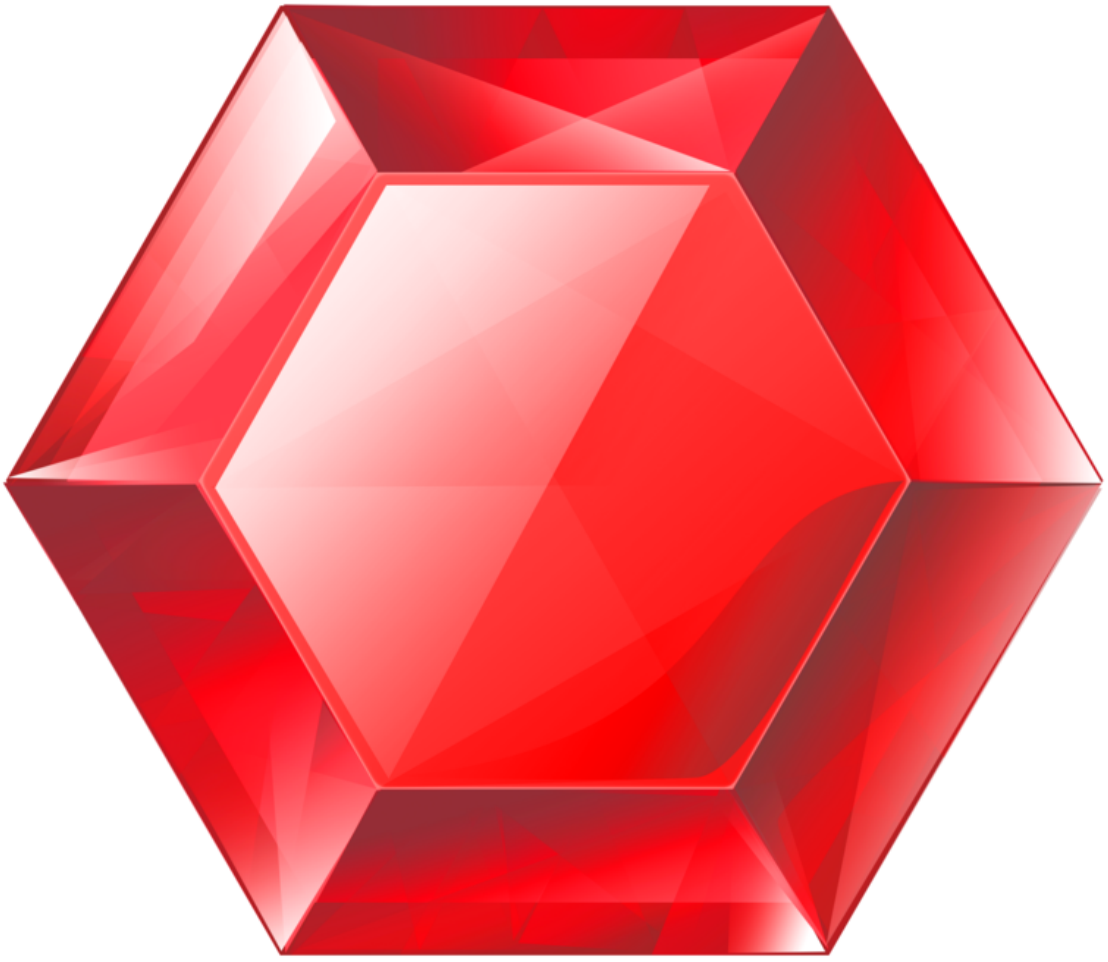
gemstone-rs Codegen Guide

Generating checked-in Rust wrappers for GemStone classes



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Rust Codegen

`gemstone-rs` codegen creates small Rust wrapper structs around GemStone OOPs. The goal is to move repeated selector strings into checked-in generated code while keeping the mapping reviewable.

Install

```
cargo install gemstone-rs-cli
gemstone-rs codegen --help
```

From a checkout:

```
cargo run -p gemstone-rs-cli -- codegen preview examples/codegen/gemstone-rs.codegen
cargo run -p gemstone-rs-cli -- codegen explain examples/codegen/gemstone-rs.codegen
cargo run -p gemstone-rs-cli -- codegen check-profile default examples/codegen/
gemstone-rs.codegen-profiles.json
```

Config Format

The config is intentionally line-oriented:

```
output = examples/codegen/generated/gemstone_wrappers.rs
class = Object
method = Object>>printString | return=String | doc=Return the receiver printString.
method = Object>>class
```

Use `Dictionary:ClassName` when a class should resolve from a specific dictionary:

```
class = UserGlobals:OkzBooking
method = UserGlobals:OkzBooking>>findById: | args=id | return=Oop | doc=Find a
booking by id.
```

Class-side references use `class`:

```
class = UserGlobals:OkzBooking class
method = UserGlobals:OkzBooking class>>findById: | args=id | return=Oop
```

Method Metadata

Option	Example	Purpose
`args`	`args=id,user`	Names generated Rust function arguments.
`return`	`return=String`	Generates a typed return helper instead of `Value`.
`doc`	`doc=Find by id.`	Writes a Rust doc comment above the generated method.

When `args` is omitted, codegen now infers useful argument names from selector keywords. For example `at:put:` becomes `at, put`, and `withCustomer:amount:` becomes `customer, amount`. Provide explicit `args=...` when the selector keywords are not good Rust argument names.

Generated wrapper files include a small `#[cfg(test)]` surface-name test stub, so downstream crates can keep generated wrapper names visible to Rust test tooling. Use `codegen explain` before generating when you want a readable summary of the output file, test stub, wrapper classes, selectors, argument names, return helpers, mapped structs, and BridgeRoot field mappings:

```
gemstone-rs codegen explain examples/codegen/gemstone-rs.codegen
```

Supported return types:

Config value	Rust return
`Value`	`gemstone_rs::Value`
`String`	`String`
`SmallInt`	`i64`
`Bool`	`bool`
`Oop`	`gemstone_rs::Oop`

Mapped Structs

Codegen can also emit typed `BridgeMapped` structs for values stored under `BridgeRoot`:

```
mapped = BookingDraft | doc=A typed Rust payload stored under BridgeRoot.
field = BookingDraft.name | type=String | key=name
field = BookingDraft.amount | type=SmallInt | key=amount | key_type=Symbol
```

```
field = BookingDraft.currency | type=String | key=currency
field = BookingDraft.tags | type=Vec<String> | key=tags
```

Supported field types are `String`, `SmallInt`, `Bool`, `Oop`, `Mapped<OtherStruct>`, and `Vec<T>`.

Field options:

Option	Example	Purpose
`type`	`type=Vec<String>`	Rust/GemStone field conversion.
`key`	`key=amount`	GemStone dictionary key.
`key_type`	`key_type=Symbol`	Use string keys or symbol keys explicitly.
`doc`	`doc=Booking amount.`	Writes a Rust doc comment above the field.

Commands

Create a starter config:

```
gemstone-rs codegen init gemstone-rs.codegen
```

Generate a config from a live stone:

```
gemstone-rs codegen discover gemstone-rs.codegen Object
gemstone-rs codegen discover gemstone-rs.codegen UserGlobals:OkzBooking
gemstone-rs codegen discover-mapping gemstone-rs.codegen BookingDraft Object
```

From a checkout:

```
cargo run -p gemstone-rs --example codegen_discover
cargo run -p gemstone-rs --example codegen_discover_mapping
```

Preview without writing:

```
gemstone-rs codegen preview gemstone-rs.codegen
```

Show a generated diff:

```
gemstone-rs codegen diff gemstone-rs.codegen
```

Check freshness in CI:

```
gemstone-rs codegen check gemstone-rs.codegen
```

Write generated wrappers:

```
gemstone-rs codegen generate gemstone-rs.codegen
```

Explorer Profiles

The local explorer can save repeatable codegen workflows as profiles. A profile captures:

- codegen root
- config path
- mapped Rust type
- GemStone class

Use the browser UI to save a local profile, export a single profile as JSON, or load a project-level profile file. The repository includes a sample:

```
examples/codegen/gemstone-rs.codegen-profiles.json
```

Validate it from CI or a terminal:

```
gemstone-rs profile sample
gemstone-rs profile init gemstone-rs.codegen-profiles.json
gemstone-rs profile validate gemstone-rs.codegen-profiles.json
gemstone-rs profile validate --json gemstone-rs.codegen-profiles.json
gemstone-rs profile list gemstone-rs.codegen-profiles.json
gemstone-rs profile show default gemstone-rs.codegen-profiles.json
gemstone-rs profile resolve default gemstone-rs.codegen-profiles.json
gemstone-rs profile check gemstone-rs.codegen-profiles.json
gemstone-rs profile check --json gemstone-rs.codegen-profiles.json
gemstone-rs codegen preview-profile default gemstone-rs.codegen-profiles.json
gemstone-rs codegen diff-profile default gemstone-rs.codegen-profiles.json
gemstone-rs codegen check-profile default gemstone-rs.codegen-profiles.json
gemstone-rs codegen generate-profile default gemstone-rs.codegen-profiles.json
```

See Codegen Profile Schema and `schemas/gemstone-rs.codegen-profiles.schema.json` for editor validation.

The project profile file uses this shape:

```
{"kind":"gemstone-rs-explorer-codegen-profiles","version":1,"profiles":
[{"name":"default","config":"examples/codegen/gemstone-
rs.codegen","root":"","mapped":"BookingDraft","className":"Object"}]}
```

Start the explorer with a project root:

```
gemstone-rs-explorer --port 8787 --codegen-root /path/to/gemstone-rs
```

Profile writes are disabled unless the explorer was started with `--allow-write`. The server rejects path traversal in `config=` and `profile_file=` after URL decoding, rejects traversal in `root=`, keeps writes under the configured codegen root by default, and requires `--allow-absolute-write-paths` before absolute write targets are accepted. Project profile saves reject unsupported top-level fields, unsupported profile fields, missing or duplicate profile names, and non-string profile values.

Generated Shape

For:

```
method = Object>>printString | return=String | doc=Return the receiver printString.
```

The generated wrapper includes:

```
pub fn print_string(&mut self) -> Result<String> {
    let value = self.session.perform(self.oop, "printString", &[])?;
    match value {
        Value::String(value) => Ok(value),
        Value::Oop(oop) => self.session.fetch_string(oop),
        other => Err(Error::UnexpectedType {
            expected: "String",
            actual: format!("{other:?}"),
        }),
    }
}
```

Generated files are meant to be checked in. That makes diffs, reviews, and editor indexing predictable.

Generated Wrapper App

The repository includes a small live example that imports the checked-in generated wrapper and calls `Object>>printString` on a small integer:

```
cargo run -p gemstone-rs --example generated_wrapper_app
```

Expected output:

```
generated wrapper printString: 7
```

The example uses the generated `Object::from_oop` constructor:

```
#[path = "../../../examples/codegen/generated/gemstone_wrappers.rs"]
mod gemstone_wrappers;

use gemstone_rs::{Config, Session};

fn main() -> gemstone_rs::Result<()> {
    let mut session = Session::login(Config::from_env()?)?;
    let oop = session.smallint_oop(7);
    let mut object = gemstone_wrappers::Object::from_oop(&mut session, oop);
    println!("{}", object.print_string()?);
    Ok(())
}
```

Generated Object Mapping

The generated file can also include Rust data structs that implement `BridgeMapped`. Run:

```
cargo run -p gemstone-rs --example generated_mapping_app
```

Expected output:

```
generated mapped payload: BookingDraft { name: "Tariq", amount: 100, currency: "GBP",
tags: ["priority", "demo"] }
```

The config-driven mapping emits a struct like:

```
#[derive(Clone, Debug, Eq, PartialEq)]
pub struct BookingDraft {
    pub name: String,
    pub amount: i64,
    pub currency: String,
    pub tags: Vec<String>,
}
```

and an implementation of `BridgeMapped` so it works with:

```
bridge_root.put_mapped("GeneratedBookingDraft", &draft)?;
let loaded: BookingDraft = bridge_root.get_mapped("GeneratedBookingDraft")?;
```

VS Code Workflow

The `gemstone-rs Workbench` extension exposes the same flow in the GemStone RS sidebar:

- Discover from Live Stone
- Generate Mapping Config
- Preview BridgeRoot
- Run Generated Mapping Example
- Preview Wrappers
- Diff Generated Output
- Check Freshness
- Generate Wrappers
- Load Project Profiles
- Save Project Profiles
- Export Codegen Profile
- Validate Project Profiles
- Check Project Profiles
- Open Codegen Docs

`Generate Wrappers` shows the diff first and asks before writing when output would change.

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